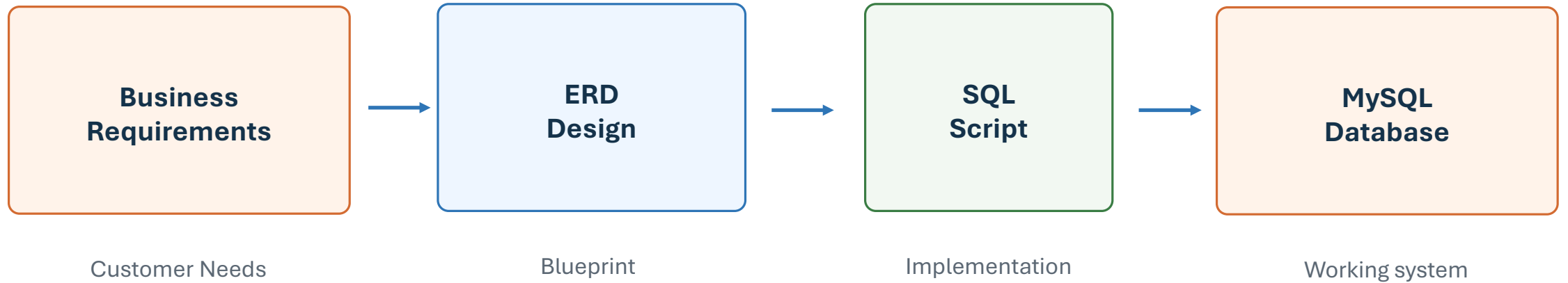


Writing SQL Using DDL, DML, and DQL

Where SQL Fits in the Database Process

SQL is where the database design becomes something MySQL can build and run.



What is an SQL Script?

An **SQL script** is a text file or block of text that contains **one or more SQL statements** that can be executed in sequence by a database system such as MySQL.

An SQL script can contain commands for:

1. **Creating or modifying database structures** — CREATE, ALTER, DROP
2. **Adding or changing data** — INSERT, UPDATE, DELETE
3. **Retrieving data** — SELECT
4. **Creating constraints, views, procedures, triggers**, and other database objects

SQL scripts are commonly saved as files with the **.sql** extension.

An **SQL statement** is a single complete instruction given to a database.

For example:

```
SELECT * FROM Student;
```

That is one SQL statement because it tells MySQL to retrieve all rows from the Student table.

Common SQL syntax rules include:

1. SQL statements are made from keywords such as SELECT, FROM, WHERE, CREATE, and INSERT.
2. Keywords must appear in the proper order.
3. Strings with spaces are enclosed in **single** quotation marks: 'Computer Science'.
4. A semicolon ; normally marks the end of a statement.
5. Table and column names must follow the identifier rules of the DBMS.

For **MySQL**, these are the main identifier rules/conventions:

- **Use letters, numbers, _, and \$.** Examples: Student, Student_ID, Course101.
- **Do not normally start an identifier with a number.** Prefer Room101 rather than 101Room.
- **Avoid spaces.** Prefer StudentName or Student_Name instead of Student Name.
- **Avoid special characters** such as -, /, #, @, and punctuation.
- **Do not use reserved SQL keywords** as identifiers. For example, avoid names such as SELECT, FROM, ORDER, TABLE, and PRIMARY.
- Identifiers should be **descriptive and meaningful**. StudentID is better than SID1 or x.
- MySQL generally allows identifiers up to **64 characters** for common objects such as tables and columns.
- If you absolutely need spaces, special characters, or a reserved word, MySQL lets you enclose the identifier in **backticks**:

SQL coding guidelines

These are not always required by MySQL, but they make SQL easier to read and maintain.

For example, MySQL accepts:

```
select studentid,name from student where major='Computer Science';
```

but this is easier to read:

```
SELECT StudentID, Name  
FROM Student  
WHERE Major = 'Computer Science';
```

A good SQL coding style guidelines is to:

1. Write SQL keywords in **uppercase**
2. Use consistent indentation
3. Put major clauses such as SELECT, FROM, and WHERE on separate lines
4. Use meaningful table and column names
5. End each statement with a semicolon

SQL is generally not case-sensitive for keywords. **Table names** can be case-sensitive depending on the operating system and MySQL configuration. On Windows, table names are usually not case-sensitive; on Linux, they often are.

SQL Keywords

SQL keywords are words that have a special meaning in SQL. They are part of the SQL language and tell the database what operation to perform or how a statement should be interpreted.

Examples of common SQL keywords are:

```
SELECT Name  
FROM Student  
WHERE Major = 'Computer Science';
```

They tell MySQL to:

```
SELECT — retrieve data  
FROM — specify the table  
WHERE — specify a condition
```

A **simple SQL script** is a sequence of SQL statements that performs several database tasks in order. For example, a script can create a database, select it for use, create a table, insert data, and then query that data. Here is a basic example in MySQL:

```
-- Create the database
CREATE DATABASE SchoolDB;

-- Select the database
USE SchoolDB;

-- Create a table
CREATE TABLE Student (
  StudentID INT PRIMARY KEY,
  Name VARCHAR(50),
  Major VARCHAR(50)
);

-- Populate the table with data
INSERT INTO Student (StudentID, Name, Major)
VALUES
(1, 'Alice Johnson', 'Computer Science'),
(2, 'Brian Smith', 'Biology'),
(3, 'Carla Davis', 'Business');

-- Perform a simple query
```

The script performs these steps in sequence:

- 1. CREATE DATABASE SchoolDB; creates a new database named SchoolDB.
- 2. USE SchoolDB; tells MySQL that the following statements should be executed in that database.
- 3. CREATE TABLE Student creates a table with three attributes: StudentID, Name, and Major.
- 4. INSERT INTO adds three student records to the table.
- 5. SELECT * FROM Student; retrieves and displays all rows and columns from the Student table.

The expected query result would look something like:

StudentID	Name	Major
1	Alice Johnson	Computer Science
2	Brian Smith	Biology
3	Carla Davis	Business